

Array Diagnostic test

Name (caps) _____ Student ID (9 digits)

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Lab Time _____ Desk Number: Desk-_____ (eg Desk-G02-060)

How to work out what a code fragment does: draw a table

The simplest and most reliable way to work out what a code fragment does (and to understand) is to draw a table. This is called tracing or dry running code. Let's work out what this fragment does:

```
int[]a = new int[3];      // L1
System.out.println(a[1]); // L2
a[1] = 5;                 // L3
System.out.println(a[1]); // L4
```

Executing Line L1

First a new array is created. It has 3 entries with indexes (positions) 0, 1 and 2. As this is an int array all positions in the array are initialised to 0.

LINE	a			Printed
	[0]	[1]	[2]	
L1	0	0	0	

Executing Line L2

The loop body prints what is in a[1], which is 0. Note this is NOT the first entry in the array because the numbering of positions starts from 0.

LINE	a			Printed
	[0]	[1]	[2]	
L1	0	0	0	
L2				0

Executing Line L3

The value 5 is stored in the array at position 1. Because the numbering is from 0 this is the second position (labelled 1) as shown. The other values in the array do not change.

LINE	a			Printed
	[0]	[1]	[2]	
L1	0	0	0	
L2				0
L3		5		

Executing Line L4

The loop body prints the contents of a[1], which is now 5.

LINE	a			Printed
	[0]	[1]	[2]	
L1	0	0	0	
L2				0
L3		5		
L4				5

Overall what is printed is:

0
5

You must show all your working in the following. No correct table (exactly the same as the solution)... no marks. If you do not get them all right and do not understand why, ask for help.

1. What is printed when the following code fragment has executed?

```
int[]age = new int[4];           // L1
System.out.println(age[0]);      // L2
age[0] = 18;                     // L3
age[1] = 19;                     // L4
age[2] = 21;                     // L5
age[3] = 17;                     // L6
System.out.println(age[0]);      // L7
```

Printed:

LINE	age				Printed
	[0]	[1]	[2]	[3]	

2. What is printed when the following code fragment has executed?

```
int[]age = new int[2];           // L1
age[0] = 33;                     // L2
age[1] = 27;                     // L3
age[1] = 28;                     // L4
age[0] = 42;                     // L5
System.out.println(age[1]);      // L6
```

Printed:

LINE	age		Printed
	[0]	[1]	

3. What is printed when the following code fragment has executed?

```
int[]age1 = new int[2];          // L1
int[]age2 = new int[2];          // L2
age1[0] = 60;                     // L3
age1[1] = 65;                     // L4
age2[1] = age1[0];                // L5
age2[0] = age1[1];                // L6
System.out.println(age2[1]);      // L7
```

Printed:

6

LINE	age1		age2		Printed
	[0]	[1]	[0]	[1]	

4. What is printed when the following code fragment has executed?

```
int x = 5; // L1
int[] s = new int[2]; // L2

for (int i = 0; i < 2; i++) // L3
{
    System.out.println(s[i]); // L4
    s[i] = s[i] + x; // L5
}
System.out.println(s[0]); // L6
```

Printed:

LINE	x	s		i	i < 2?	Printed
		[0]	[1]			

5. What is printed when the following code fragment has executed?

```
int[] s = new int[3]; // L1

for (int i = 2; i > 0; i--) // L2
{
    s[i] = i+4; // L3
    System.out.println(s[i]); // L4
}
```

Printed:

LINE	s			i	i > 0?	Printed
	[0]	[1]	[2]			

6. What is printed when the following code fragment has executed?

```
double[] shoe = new double[1]; // L1
System.out.println(shoe[0]); // L2
shoe[0] = 9.5; // L3
System.out.println(shoe[0]); // L4
```

Printed:

LINE	shoe	Printed
	[0]	
L1		
L2		
L3		
L4		